Lab 03: Chaotic Text Generator. One-page report

1. Problem and solution

- Text generation samples new sentences from a probabilistic model, but a generated sentence has no reference answer, so quality has to be measured on hundreds of sentences.
- A first-order HMM only sees the previous tag, so after a NOUN it cannot tell a subject (a verb follows) from an object (a preposition or full stop follows).
- We build a POS-HMM (hidden tags, word emissions) estimated by counting, with temperature sampling and Viterbi tagging in log space.
- We add role-split states (S / O / P) so the model knows where it is in the sentence, and compare with a trigram baseline on the same data.
- A Streamlit test bench (app . py) generates batches, shows hidden tags, computes four metrics, tags typed sentences and exports CSV.

2. Metrics at T = 1.0 (500 sentences, seed 42)

Model	Valid tag pattern	Plausible subject-verb	Novel sentences	Distinct-2
HMM, plain states	16.2%	14.8%	99.8%	0.198
HMM, role-split states	100%	60.6%	99.4%	0.148
Trigram	57.0%	76.0%	91.2%	0.105

The plain HMM scores low because after a NOUN its next tag is VERB 0.39, PREP 0.22 and PUNCT 0.39 whatever the role of the noun, so it often ends a sentence after an ADJ NOUN or builds an invalid pattern. Role-split states give the model the memory of whether it is before the verb, after the verb or inside a prepositional phrase. This removes every invalid pattern (100%). The trigram remembers two previous words, so it reaches 57% on patterns, but it cannot enforce the full sentence shape.

3. Choosing a temperature

- **Writing toy: T = 1.5 (role-split HMM).** Structure stays at 100% valid and distinct-2 rises to 0.161, so the output is varied and still grammatical; surprises are welcome in a toy.
- **Autocomplete: T = 0.5 (role-split HMM).** Structure stays at 100% valid and predictions stay close to the most likely continuation; a low temperature avoids odd suggestions. The trigram should not be used here, because at T = 0.5 it copies training sentences.

4. Failure examples (T = 1.0, seed 42)

Sentence	Failure type	Cause
“the quiet school .” (plain HMM)	**Structure**: tag pattern DET ADJ NOUN PUNCT has no verb, so it is not in the 16 training patterns	First-order model: after ADJ NOUN the next tag PUNCT is possible, so the sentence can end without a verb.
“the foxes lose by the school .” (role-split HMM, from `results/failures.txt`)	**Meaning**: valid pattern, but the pair foxes-lose never occurs in training	The states encode role, not word meaning, so any noun can be paired with any verb of the same role.
“the girls leave the park .” (trigram, T = 0.5)	**Memorisation**: copied from the training set	At low temperature the trigram counts are sharp, so it repeats a seen sentence.

Good example (role-split HMM, T = 1.0, not in training): “the teachers eat the birds near the river .” It has a valid pattern and a plausible subject-verb pair, but it is only fluent, not meaningful: the model has no idea of what teachers or birds are.

5. What state splitting changed, and what it did not

State splitting changed the structure of the output: valid tag patterns rose from 16.2% to 100%, and plausible subject-verb pairs rose from 14.8% to 60.6%, because the verb is now chosen with the subject role in mind. It also made the hidden states more readable (DET_S, NOUN_O, PREP ...). It did not change the vocabulary, so word choices are still arbitrary and meaning is not modelled. Novelty stayed near 99%, while distinct-2 fell from 0.198 to 0.148 (fewer distinct word pairs, since the structure is now more constrained). Plausible subject-verb (60.6%) is still below the trigram (76.0%). Tagging accuracy on the synthetic corpus was 100% both before and after the split, because the grammar is unambiguous once the previous tag is known.

6. Synthetic versus real text

Model	Accuracy	Known words	Unknown words	Whole sentence correct
Most-frequent tag (baseline)	86.2%	90.8%	28.6%	31.9%
HMM + Viterbi	88.1%	92.9%	28.2%	40.4%
Synthetic corpus, HMM + Viterbi	*100%*	*100%*	*n/a (no unseen words)*	*100%*

The HMM gains only 1.9 points over the most-frequent-tag baseline on English EWT (17 tags, 16,654-word vocabulary, 10,359 distinct tag sequences for 12,544 sentences). Unknown words are 7.5% of test tokens and are tagged at only 28.2% accuracy; errors go to rare tags such as X and INTJ (PROPN tagged X 195 times, INTJ 137 times). The synthetic corpus has 6 tags, 61 words and only 16 tag sequences, so Viterbi is trivially right there. Real text has a large vocabulary, many unseen words and almost one tag sequence per sentence, so the synthetic results do not transfer. The valid-tag-pattern metric is also meaningless on real text, since it is defined relative to 16 synthetic sequences.